Verification of Probing Security for Masked Circuits

A Constructive Approach and Theoretical Study

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5. It verifies a suite of masked gadgets exactly, and the coupling view proves to be helpful.

Motivation & Model

Side-channel attacks

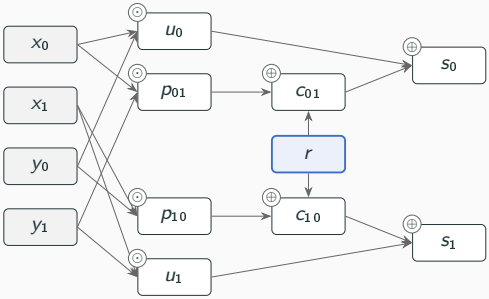
A cipher can be secure on paper and insecure on the bench.

- A real device leaks through power draw, timing, electromagnetic emanation; all of which vary with the *data* it handles.
- Differential power analysis (Kocher et al. 1999) correlates many power traces against a guess of an intermediate value, and may recover the key.

So, the insecurity lies not in the mathematics behind the cipher, but in the way the hardware computes it.

Masking — and our running example

Boolean masking (Chari et al. 1999) splits each secret into shares, $x = x_0 \oplus x_1$, with x_0 uniformly random, and computation happens on shares, never directly on secrets. Then, the cost of attacking grows exponentially in the number of shares.



DOM-AND (Groß et al. 2016): a masked AND gate, with two shares per input, and a fresh mask r.

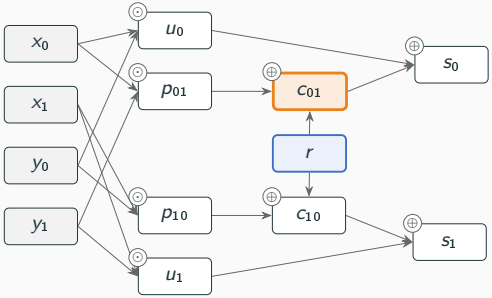
- computes a sharing $s_0 \oplus s_1 = x \odot y$,
- the cross terms p_{01}, p_{10} mix shares of *both* secrets — the delicate part,
- r re-masks them before they meet an output.

This gadget accompanies us for the rest of the talk.

The probing model

Reasoning about noisy traces directly is unwieldy. Luckily we have an abstraction (Ishai et al. 2003):

d-probing security
An adversary picks any d wires and is handed their *exact* values. The circuit is d-probing secure when no such choice distinguishes one secret assignment from another.



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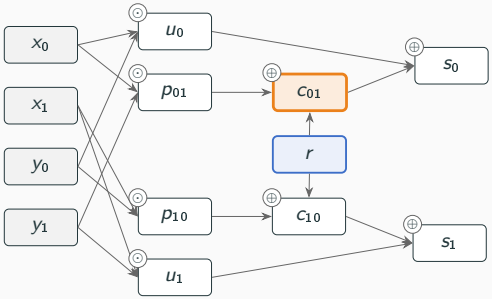

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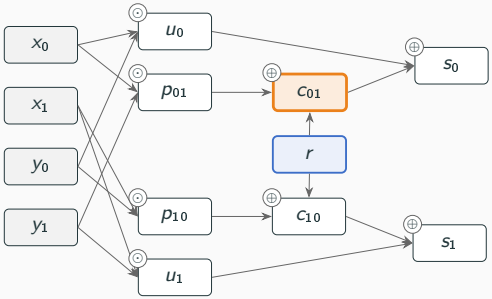

- Now, security becomes a *combinatorial* property of the circuit, algortihmically decidable in principle.
- And thankfully, the abstraction is faithful: probing security implies noisy-leakage security (Duc et al. 2014). We take d-probing security as the target and do not revisit this bridge.

The problem

Setting. We work with arithmetic circuits over $\mathbb{F}_2$, two-input $\oplus$ and $\odot$ gates, where inputs are partitioned into secrets $\mathcal{S}$, randoms $\mathcal{R}$, publics $\mathcal{P}$. We have value-based leakage, so, no glitches, and also no composition.

Exact verification of d-probing security
One should decide, for *every* one of the $\sum_{i \leq d} \binom{|\mathcal{W}|}{i}$ observation sets, whether it is independent of the secrets.

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- Two difficulties arise: a *single* probe already asks a question about a distribution over all of $\{0,1\}^{\mathcal{R}}$,
- and there are combinatorially many probes to settle.

How Hard Is One Probe?

Warm-up: three views of secret-independence

Fix a probe O (a tuple w of wires). Write $\mathcal{L}_{\mathsf{a}}(O)$ for its distribution over a uniform tape $\rho \in \{0,1\}^{\mathcal{R}}$, and $N_{\mathsf{a}}(o) = |\{\rho : O(\rho; \mathsf{a}) = o\}|$ for the tape counts.

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All three are equivalent — but they are *not* equally usable. The counting view tells us what a verification procedure should not attempt directly; the coupling view tells us what it might be able to acomplish.

Counting is intractable

Theorem (thesis, Ch. 4)
Computing one tape count $N_{\mathsf{a}}(o)$ is $\#\mathrm{P}$-complete (Valiant 1979). And deciding $\mathcal{L}_{\mathsf{a}}(O) = \mathcal{L}_{\mathsf{b}}(O)$ is $\mathrm{C}_{=}\mathrm{P}$-complete (Wagner 1986).

Hardness, briefly: for formulas φ_0, φ_1 over randoms $x_1, \ldots, x_m$ and a single secret bit s, the wire

$$w \;=\; \varphi_0 \oplus \big(s \odot (\varphi_0 \oplus \varphi_1)\big)$$

reads φ_0 when $s = 0$ and φ_1 when $s = 1$. Then, the one-bit probe $\{w\}$ is secret-independent *iff* $\#\varphi_0 = \#\varphi_1$. That is exactly the canonical $\mathrm{C}_{=}\mathrm{P}$-complete question: ExactSat.

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One secret bit and one probed wire already suffice for the problem to be intractable; no feature of masking is what makes the problem hard.

What the placement tells

- Observe that $\mathrm{C}_{=}\mathrm{P} \subseteq \mathrm{P}^{\#\mathrm{P}}$, so the decision never costs more than the counting. Another observation is the following: a formula is unsatisfiable exactly when its count is zero, i.e. $\mathrm{coNP} \subseteq \mathrm{C}_{=}\mathrm{P}$. Hence, no polynomial exact decision exists unless $\mathrm{P} = \mathrm{NP}$.

Consequence

We do not compute or compare the counts. We certify security *constructively*: by exhibiting, for each probe, a measure-preserving relabelling of the randomness under which the secrets visibly cancel. A witness proves independence without explicitly counting — at the price of completeness!

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- Both classes sit *above the polynomial hierarchy* (due to Toda). A single probe is already there, and full verification universally quantifies over exponentially many such instances.

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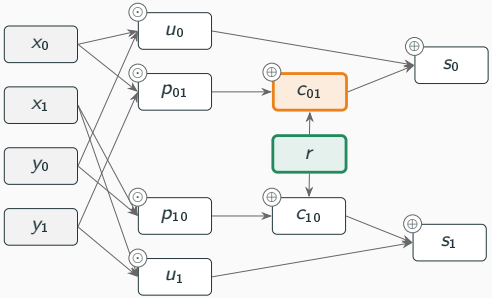
Security Is a Coupling

One substitution, on the running example

Probe the masked cross term: $c_{01} = x_0 \odot y_1 \oplus r$.



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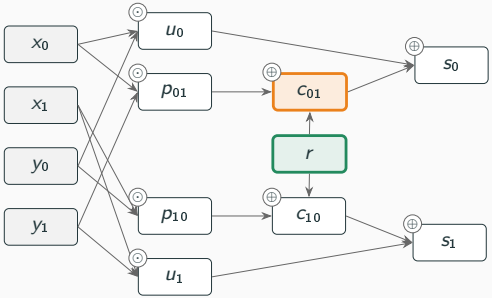

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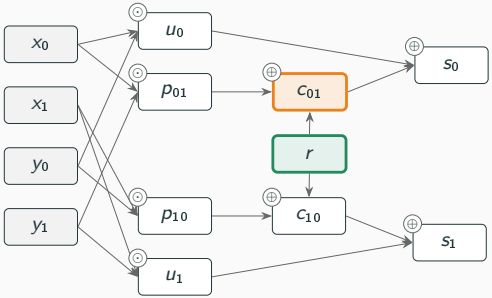

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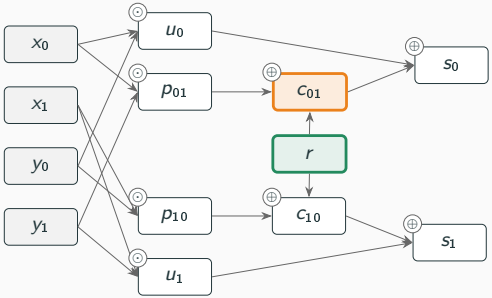

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Key identity: $(\mathrm{w}\sigma)(\rho) = \mathrm{w}(\widehat{\sigma}(\rho))$ — the syntactic step and the semantic relabelling agree.

Couplings

The classical idea ((Barthe et al. 2017) for probabilistic programs) to prove that two distributions are equal without explicitly computing them is by *linking their sources*. Our two laws are pushforwards of the *same* uniform tape. So we should cleverly link those tapes.

Definition (coupling function)
A *coupling function* from a to b for a probe O is a bijection $\varphi : \{0,1\}^{\mathcal{R}} \to \{0,1\}^{\mathcal{R}}$ with

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- A bijection of a finite set preserves the uniform law trivially. The identity says that a run under a is reproduced, observation for observation, under b. This is a *diagonal* coupling of the two laws. It proves that the distributions on the observation set are the same for those two secrets.

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Theorem
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- ($\Rightarrow$) Sort the tapes by outcome: $F_{\mathsf{a}}(o) = \{\rho : O(\rho; \mathsf{a}) = o\}$. As o varies these *fibres* partition $\{0,1\}^{\mathcal{R}}$, and equal laws mean matching fibres have equal size, $|F_{\mathsf{a}}(o)| = |F_{\mathsf{b}}(o)|$. So pick, for each o, any bijection $F_{\mathsf{a}}(o) \to F_{\mathsf{b}}(o)$; their union is a bijection φ of the whole tape space, and it sends every ρ to a tape with the *same* observation, giving us a coupling function. □

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This is an existence statement and not a procedure
The ($\Rightarrow$) direction matches fibres *block by block*, and for that it needs the counts $N_{\mathsf{a}}(o)$, which are #P-hard to compute. So, the theorem guarantees a coupling, it does not find one.

From substitutions to couplings — the constructive route

- Each substitution induces a shear, a bijection of the tapes. A *sequence* of substitutions composes to a bijection φ_{a} — one per secret assignment, since the contexts e_i may mention secrets.

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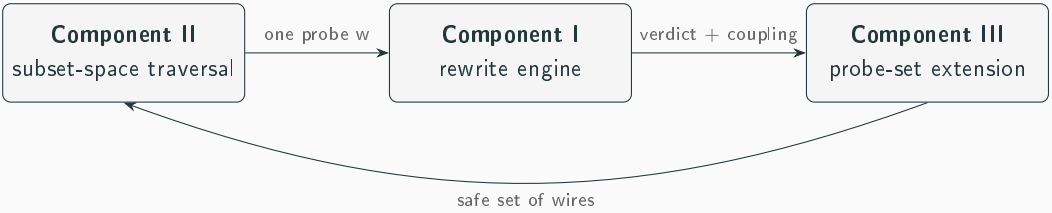
- So a successful rewrite derivation is not a simple Secure verdict. It is an explicit, checkable, reusable *certificate*.

This is the design principle of Coppula: replace intractable counting by the search for a coupling, building it one substitution at a time.

Coppula: Three Components

Pipeline overview



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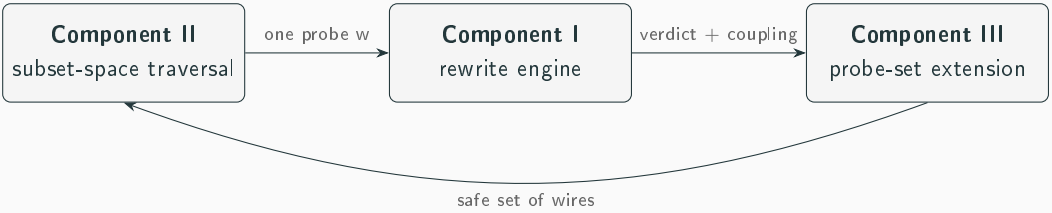

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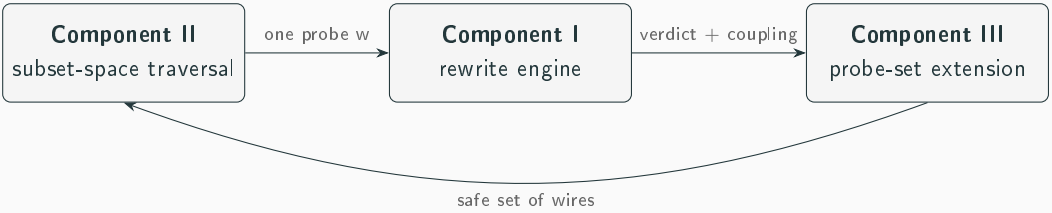

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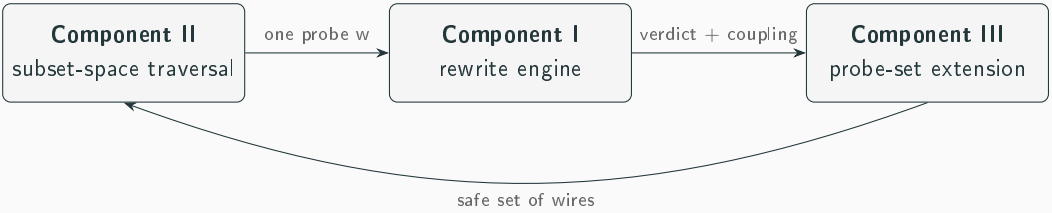

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- II sweeps *every* d-subset without explicitly enumerating them.
- III amplifies one certificate into a whole *safe set*, shrinking the space II must sweep.

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Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

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- **Sound.** Every step is a shear, so any Secure verdict carries a genuine coupling (Theorem, Ch. 5).
- **Incomplete, knowingly.** A probe can be independent for reasons no relabelling of this kind captures — this is the price of the hardness result.

Component I: the rewrite engine

Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

- **Tiered.** A cheap single-occurrence rule first (r occurs once across w: substitute, done); with a complete substitution loop behind it.
- **Sound.** Every step is a shear, so any Secure verdict carries a genuine coupling (Theorem, Ch. 5).
- **Incomplete, knowingly.** A probe can be independent for reasons no relabelling of this kind captures — this is the price of the hardness result.

On the example: $c_{01} \xrightarrow{r \mapsto a_0 \odot b_1 \oplus r} r$ is a single step discharge, which is Secure, and records a coupling.

Component I: why factoring

Some masks are buried inside *products*, where substitution cannot reach them.

- Substituting $r \mapsto e \oplus r$ needs r to occur as a *summand*; in $u \odot r \oplus u \odot v$ every occurrence of r is trapped inside a product.

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  and the hidden random surfaces as a summand again, ready to substitute.
- Factoring is applied lazily, on demand, between substitution attempts; the engine interleaves the two until the probe is discharged or no rule applies.

Component II: covering all probes

Certifying each d-subset in turn is infeasible. Instead the space lives as a *worklist* of factors $\langle (c_1, W_1), \ldots, (c_m, W_m) \rangle$, "draw c_j wires from each W_j", and we split recursively, in the manner of maskVerif (Barthe et al. 2019):

- Initially, try to discharge a whole batch at once: if the *union* of its wires is jointly secret-free, every probe inside is also settled — prune the branch.

Theorem (exhaustiveness, Ch. 5)
The recursion visits every d-subset exactly once. The proof reads Vandermonde's identity $\binom{m+n}{d} = \sum_k \binom{m}{k}\binom{n}{d-k}$ at the level of *sets*: the cells genuinely partition the probes, they do not merely count them.

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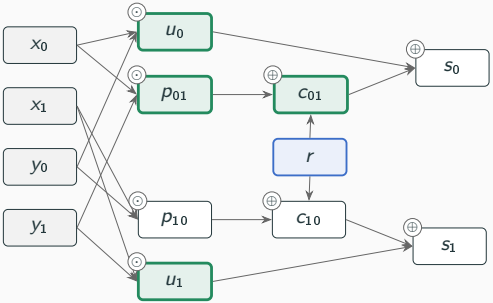
The delicate point is that the split must *lose nothing*, and we proved that it does not.

Component III: growing safe sets

A discharged probe is a coupling — but can the *same* coupling cover more wires?



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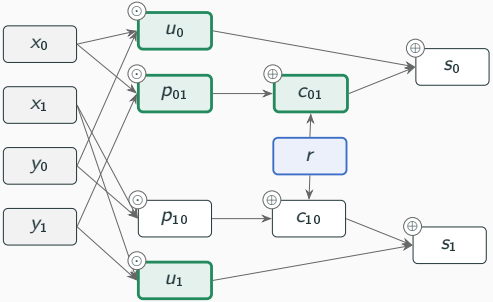

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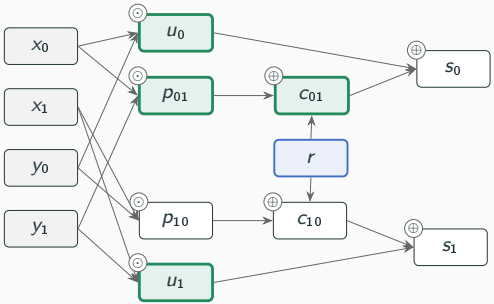

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- Exact test: a function depends on a variable iff the variable occurs in its *ANF* — so the endpoint test is to send expressions into their normal froms.
- One certificate, with more wires: the whole safe set is settled by a single discharge, and Component II skips it.

Component III: three mechanisms

- **Closure** — build-time, entropy-style deduction ($H(u \mid \mathrm{w}) = 0$). Works locally, but the candidate pool deep in the recursion is scattered. A cheap step but weak.

Proposition (nesting, Ch. 5)
closure $\subseteq$ replay $\subseteq$ coupling — every wire the cheaper mechanisms admit, the coupling admits too. Coppula adopts the coupling approach.

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- **Coupling** — apply the recorded tape bijection and test the ANF endpoint. Exact, and independent of how the derivation was found.

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Take the certified probe w with coupling φ, and a candidate wire u from the current pool. Decide whether $u \circ \varphi$ secret-free?

Phase 1 — fast reject (probabilistic, may only *reject*)
Sample tapes; evaluate $u \circ \varphi$ twice per tape, secrets drawn at random vs. all zero. A disagreement proves $u \circ \varphi$ moves with the secrets: drop u. Agreement proves nothing, survivors move on.

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Why blinded wires are safe to add. For an admitted u, $u(\varphi_{\mathsf{a}}(\rho); \mathsf{a}) = \overline{u}(\rho)$, a function of ρ alone. Since φ_{a} preserves the uniform law, the *joint* law of all admitted wires under a is that of $(\overline{u}(\rho))_u$, which mentions no secret (Theorem, Ch. 5).

Implementation & Evaluation

Implementation details

Coppula is implemented in **Lean 4**.

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Design and proofs are implementation-agnostic; these choices simply make algorithms fast.

Evaluation: outcomes

Forty configurations at orders 1–4: ISW multiplications, DOM and TI ANDs, refresh gadgets, Keccak χ (DOM and TI sharings), and the TI S-boxes and quadratic sharings of Bilgin et al. Every verdict matches the ground truth, and every Insecure comes with a concrete witness observation.

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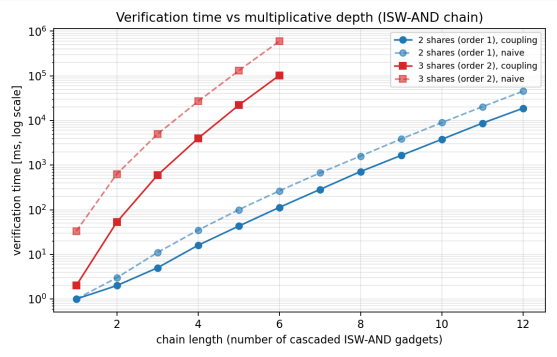
(Full extension-comparison table @ the appendix slides.)

Evaluation: is the extension worth it?



Verification time versus circuit depth.

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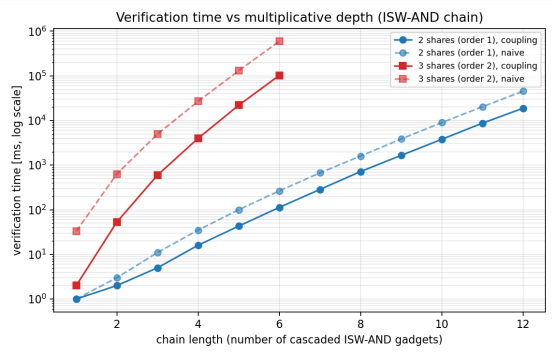

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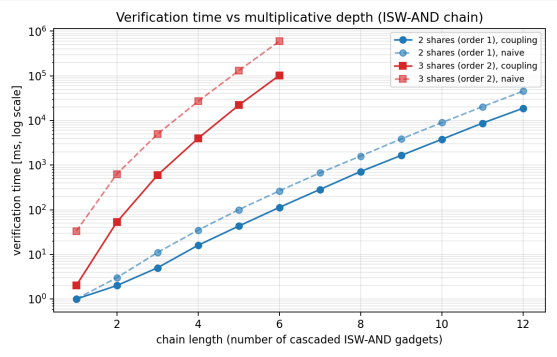

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- The extension pays for itself: Fides certifies 537 wires from 55 discharges; five-share ISW falls from ∼609 s (no extension) to under 2 s.

Wrap-up

Limitations and future work

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- **More.** smarter factoring heuristics; exploiting circuit symmetries to quotient the subset space.

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4. **Evaluation.** We present exact verdicts, with witnesses, across a suite of masked gadgets, and the coupling extension wins in coverage.

Thank you!

Questions?

Backup

Backup: extension comparison (I/IV)

Each entry: Coupling / Replay / Closure. Verdict S = Secure, I = Insecure; † marks replay's false-Insecures.

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
ISW multiplication					
2 sh.	1	S/S/S	0/0/1	7/7/15	2/2/0
3 sh.	2	S/S/S	3/1/34	34/34/309	31/31/8
4 sh.	3	S/S/S	46/37/3190	338/338/10770	437/437/439
5 sh.	4	S/S/S	1125/845/409472	4735/4735/632834	8078/8078/37513
DOM AND					
2 sh.	1	S/S/S	1/0/0	11/11/15	2/2/0
2 sh.	2	I/I/I	0/0/1	2/2/6	0/0/2
3 sh.	2	S/S/S	5/4/35	88/88/325	44/44/7
4 sh.	3	S/S/S	104/90/3419	859/859/13291	784/784/532

Backup: extension comparison (II/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
Trichina AND					
well-formed	1	S/S/S	0/1/1	11/11/15	2/2/0
mis-associated	1	I/I/I	1/0/1	10/10/12	1/1/0
TI AND					
3 sh.	1	S/S/S	0/0/1	9/9/27	9/9/0
3 sh.	2	I/I/I	1/1/1	10/10/10	5/5/2
Mask refreshing					
additive, 3 sh.	2	S/S/S	0/0/0	6/6/8	1/1/0
additive, 4 sh.	3	S/S/S	1/0/1	11/11/17	6/6/0
ISW, 3 sh.	2	S/S/S	0/0/0	6/6/10	4/4/0
ISW, 4 sh.	3	S/S/S	0/1/2	12/12/51	24/24/0
Keccak χ					
DOM, 2 sh.	1	S/S/S	14/11/29	49/49/107	29/29/5
DOM, 3 sh.	2	S/S/S	2807/1797/11563	2865/2865/14047	2671/2671/645
TI, 3 sh.	1	S/S/S	21/14/103	43/43/183	68/68/5
TI, 3 sh.	2	I/I/I	21/14/72	57/57/352	92/92/27

Backup: extension comparison (III/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
TI S-boxes and quadratic sharings					
Q_{12}^4 direct, 2 sh.	1	S/I†/S	2/2/5	27/26/47	16/16/10
Q_{12}^4 TI, 3 sh.	1	S/S/S	5/4/25	25/25/83	30/30/6
Fides AB1, 4 sh.	1	S/S/S	1941/972/36461	55/55/1127	537/537/14
$x + yz$ direct, 3 sh.	1	S/S/S	1/0/3	7/7/29	10/10/0
$x + yz$ w/ CT, 3 sh.	1	S/S/S	1/1/3	11/11/29	9/9/0
$x + yz$, 4 sh.	2	S/S/S	11/8/170	78/78/767	115/115/20
$x + yz$, 4 sh.	3	I/I/I	11/7/8	102/102/72	99/99/15
$x + yz + xyz$, 4 sh.	1	S/S/S	45/18/175	31/31/167	63/63/1
xy virtual var.	1	S/S/S	3/1/17	9/9/55	23/23/0
xy virtual share	1	S/S/S	1/0/7	9/9/37	15/15/0
xy, 4$\rightarrow$3 sh.	1	S/S/S	1/1/3	11/11/25	8/8/0

Backup: extension comparison (IV/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
$d+1$ **quadratic S-box family**					
Q_1^3	1	S/S/S	0/1/0	13/13/23	11/11/10
Q_2^3	1	S/I†/S	2/1/4	25/24/43	13/13/6
Q_3^3	1	I/I/I	4/2/7	18/18/32	22/22/2
Q_4^4	1	S/S/S	1/0/1	15/15/27	14/14/14
Q_{12}^4	1	S/I†/S	2/2/5	25/24/47	15/15/8
Q_{293}^4	1	S/I†/S	4/2/6	31/24/59	20/20/8
Q_{294}^4	1	S/S/S	1/1/3	23/23/39	16/16/12
Q_{299}^4	1	S/I†/S	5/2/13	35/18/75	26/22/8
Q_{300}^4	1	I/I/I	4/3/8	18/18/32	24/24/2

The two genuinely insecure classes, Q_3^3 and Q_{300}^4, are exactly the quadratic classes that admit no uniform three-share TI (Bilgin et al. 2015).

Backup: $\mathrm{C}_{=}\mathrm{P}$ membership, sketch

Encode disagreement of the two laws as a single gap function:

$$f(C, O, \mathrm{a}, \mathrm{b}) \;=\; \sum_{o} \bigl(N_{\mathrm{a}}(o) - N_{\mathrm{b}}(o)\bigr)^{2}.$$

Each N is a $\#\mathrm{P}$ function; GapP is closed under subtraction and multiplication (Fenner et al. 1994); the sum has $2^{k} \le 2^{d}$ terms, constant for fixed d. A sum of squares vanishes iff every term does, so

$$\mathcal{L}_{\mathrm{a}}(O) = \mathcal{L}_{\mathrm{b}}(O) \iff f = 0,$$

and the problem is the vanishing set of a gap function, which is exactly $\mathrm{C}_{=}\mathrm{P}$.

Backup: replay vs. factoring

The derivations of Barthe et al. (2015) replay two rule kinds: OPT (substitution) and CONV (ANF normalisation) — for them, ANF is a *replayed rule*.

Our derivations interleave substitutions with *multiplicative factoring*, and a factoring step has no tape-level counterpart to replay. Replaying the substitution steps alone on a new wire can spuriously fail, producing false-Insecure verdicts (observed in the evaluation, the † rows).

Hence the coupling mechanism: apply the recorded bijection $\widehat{\sigma}_1 \circ \cdots \circ \widehat{\sigma}_k$ semantically and test the ANF endpoint — exact, and indifferent to how the derivation was found.

Backup: positioning

- **maskVerif** (Barthe et al. 2019): language-based, the traversal's main inspiration; replay-style extension; no factoring.
- **REBECCA / Fourier-analytic** (Bloem et al. 2018): correlation sets over Fourier coefficients; approximate in general.
- **SILVER** (Knichel et al. 2020): exact, BDD-based model counting; exhaustive per-probe distributions, different scaling profile.
- **Coppula**: exact like SILVER, constructive like maskVerif — and the certificate (the coupling) is itself an object with structure we exploit.

Backup: the four-share TI netlist

Four-share TI of $x \oplus (y \odot z)$ (Bilgin et al. 2015); each input four-shared, sixteen cross products $y_i \odot z_j$:

$$F_1 = x_2 \oplus \bigoplus_{i,j \in \{2,3,4\}} y_i z_j, \quad F_2 = x_3 \oplus y_1 z_3 \oplus y_1 z_4 \oplus y_3 z_1 \oplus y_4 z_1 \oplus y_1 z_1,$$
$$F_3 = x_4 \oplus y_1 z_2 \oplus y_2 z_1, \qquad F_4 = x_1.$$

A *first-order* threshold implementation — never claimed beyond order one. Coppula certifies orders 1 and 2 and finds the order-3 witness: a counterexample on a gadget with a *known* failure order, not a flaw.

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